

Introduction to Coding and Coding Blocks

CHAPTER

6



Learning Objectives

- Introduction to Coding.
- Installation of Coding App.
- Algorithm and Flowchart.
- Areas in Scratch.
- Use of Coding Blocks.
- Variables and Operator Block.

"The computer was born to solve problems that did not exist before." — Bill Gates

Welcome to the exciting world of coding! This book is your key to unlocking the mysteries of computers, robots, and digital wonders. Have you ever wondered how your favorite video games are created, or how those funny cat videos magically appear on your screen? Well, guess what? It's all thanks to coding!

1. Introduction to Coding

What is coding?

Block-based coding is like playing with digital building blocks. Instead of stacking physical blocks to create something, you'll be stacking colorful, digital blocks on your computer screen to make amazing things happen. It's like giving instructions to a computer using puzzle pieces.

2. Installing Software

Go to www.mblock.cc



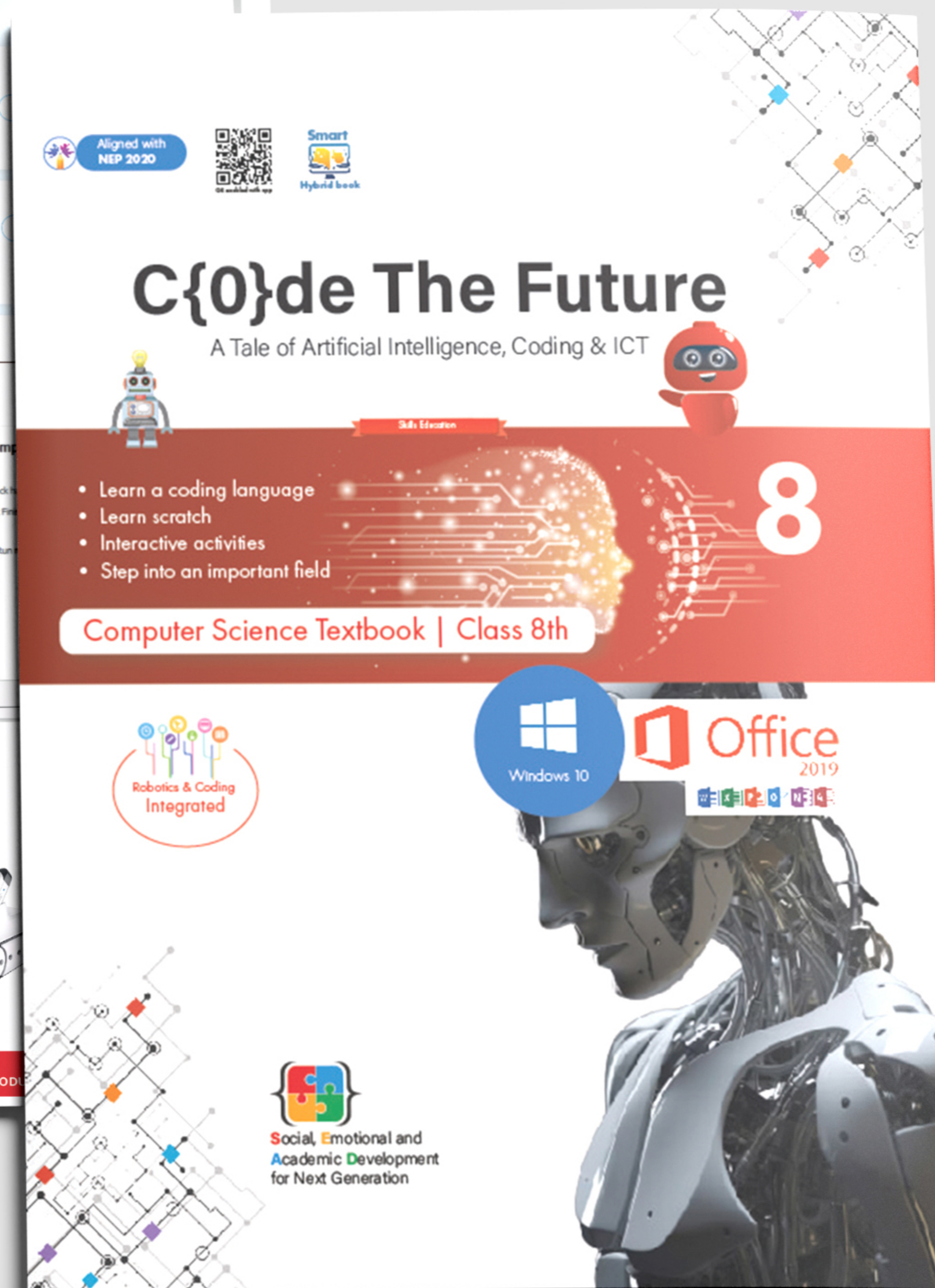
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Adding new Sprite in the Program.

Click on **Add** to choose a sprite

Click on category

Click on the sprite you like

We choose **Bat1** for this tutorial

Sprite is added

makeblock | mBlock

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4. Background Area.

The Background Area in mBlock is like the stage for your digital adventure. Just like a play or a movie has a background to set the scene, your digital project needs one too. Here, you choose and create the setting for your story or game. You can pick space backgrounds, under the sea, or a magical forest. Your characters or objects, called sprites, perform like the stage where your digital adventure comes to life, making you exciting!

Adding Background in mBlock

Upload

Background Library

My Backgrounds

The **background** will appear.

Lets know more

Block based programming was build by MIT to teach kids to code and involved in the field at a early stage which led to many great innovation in future **mBlock** is based on another block-based coding tool called Scratch.

Scratch

MIT

Massachusetts Institute of Technology

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- Learn a coding language
- Learn scratch
- Interactive activities
- Step into an important field

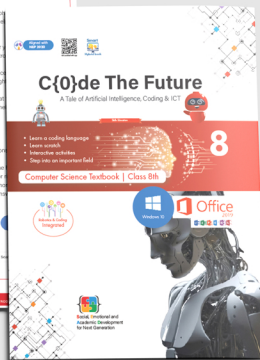
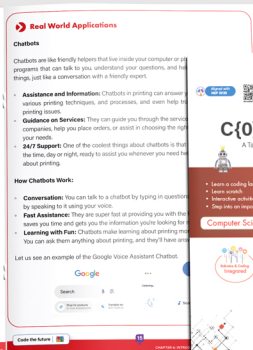
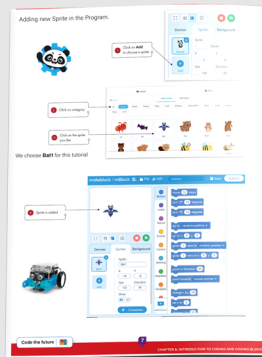
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Exercise

Fill in the Blanks

(Algorithm, Stage, Background, Operator, Block-based)

- _____ coding is like playing with digital building blocks.
- An _____ is a set of clear and precise instructions for solving a problem or completing a task.
- The _____ Area is like a magical world inside your computer.
- The _____ Area is like the stage for your digital adventure.
- _____ blocks are fundamental tools used for performing mathematical and logical operations.

Write T/F for each statement

- Block-based coding is like playing with digital building blocks. ☐
- Event blocks are used to make your code respond to actions like clicking the green flag. ☐
- The Background Area is like the stage for your digital adventure. ☐
- In mathematics, a variable is a symbol or letter used to represent an unknown or changing value in equations, functions, or mathematical expressions. ☐
- The Stop & Go buttons in mBlock allow you to control the movement of sprites in real-time during the program execution. ☐

MCQ's use tick (✓) for correct answer

- Which of the following is not a part of the mBlock platform?

☐ Spreadsheet program	☐ Presentation software
☐ Word-processing program	☐ Graphic design tool
- What does the Stage represent in mBlock?

☐ Accessing the internet	☐ Quickly accessing frequently used tools
☐ Organising files	☐ Printing documents
- What is used as a symbol or letter used to represent an unknown or changing value in equations, functions, or mathematical expressions?

☐ Quick access toolbar	☐ Saving a document
☐ Worksheet	☐ Opening a document
- Which block is used to play sound?

☐ Operator Block	☐ Motion Block
☐ Sound Block	☐ Event Block

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- Which area is like the stage for your digital adventure?

- ☐ Sprite Area
☐ Background Area

- ☐ Event
☐ Block

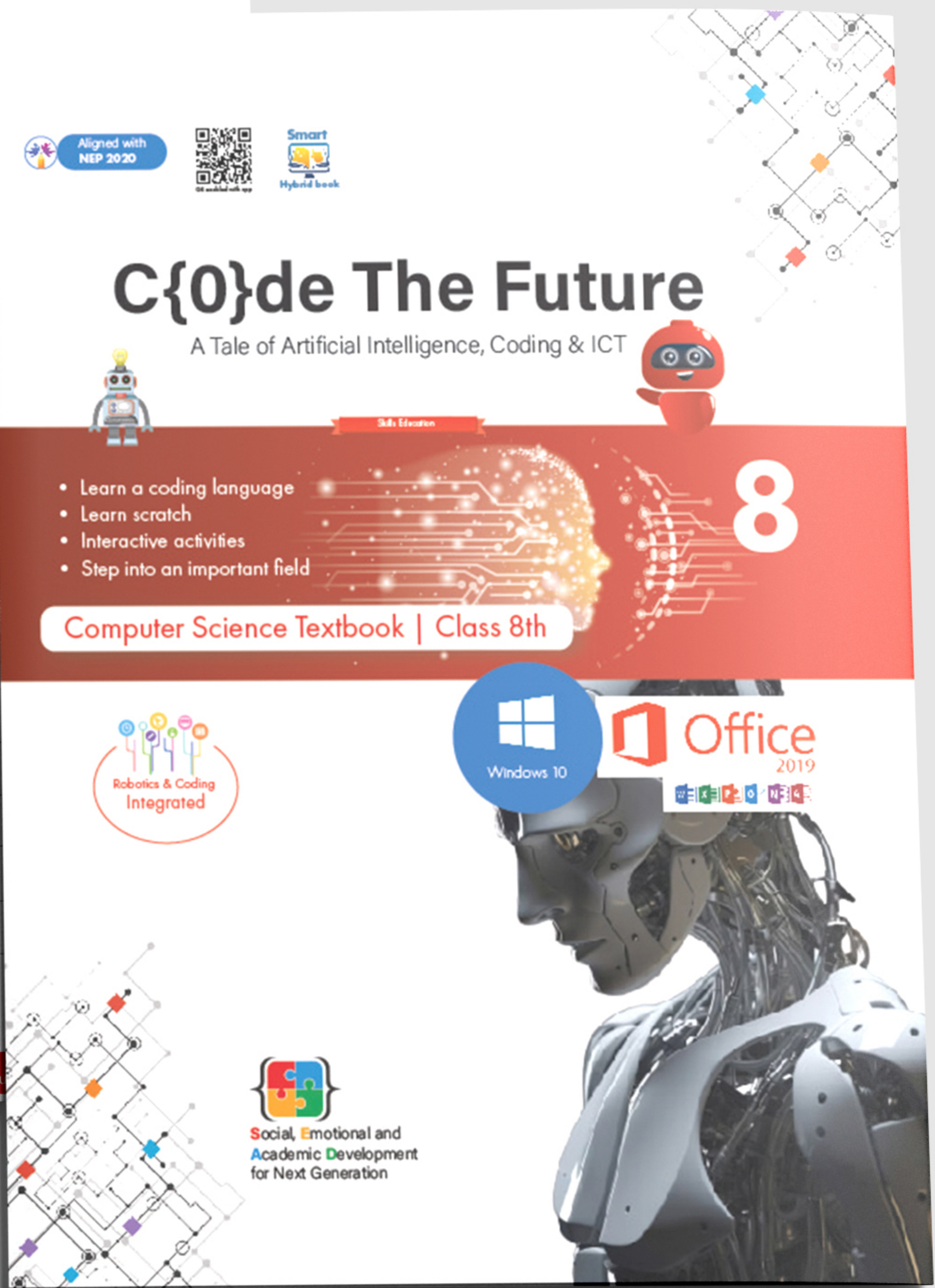
Subjective question

- What is Coding?
- What is an Algorithm?

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CHAPTER 6: INTRO



Hands on

INTEGRATED ACTIVITIES SECTION

Lab session

Let's Practice

A. Start your lab computer. Open or Install mBlock. (Ask teacher for help in installation).

1. Imagine you and your friend waiting for a school bus.

2. Add appropriate background and sprite.

Group discussion

Let's Express

A. Discuss main differences and similarities between Scratch and mBlock and why is mBlock a bit more advance and easier to use?

Project work

Let's Enjoy

A. Continue with your lab session

1. Imagine yourself and your friends sitting in a classroom.

2. Create a mBlock screen.

3. Add more classmates or characters sprites in the environment like a cat.

Teachers corner

Guided Teaching

1. Discuss what coding is and its importance.

2. Explain that coding is giving step-by-step instructions to a computer to perform tasks.

3. Show examples of everyday instructions (e.g., making a sandwich) to relate to the concept of giving precise instructions.

4. Present more complex challenges to encourage critical thinking.

5. Encourage students to create longer sequences of instructions using coding blocks to solve the challenges.

Learn more

seadnxt.com/ctf/grade3ch2

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CHAPTER 2: WORD PROCESSOR

TRIVIA

1. The first programmer is often considered to be Ada Lovelace, who created Charles Babbage's early mechanical general-purpose computer, the Analytical Engine, in the mid-1800s.

2. Coding is like learning a new language; it has syntax and grammar that you need to understand and execute.

3. Coding allows you to create and design almost anything - from websites to games and animations.

4. Coding blocks, such as Scratch and Blockly, use visual elements to make coding more accessible and fun for young learners.

Career corner

Software Developer/Engineer:

Design, develop and test software applications. They work on various platforms, including mobile, web and desktop applications.

Web Developer:

Focus on creating websites by coding HTML, CSS, and JavaScript. They may specialize in front-end or back-end development (server-side operations).

Game Developer:

Create and develop video games and specialized software.

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CHAPTER 3: GRAPHICS

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